

MCQ Exam Preparation Assistant

Mini-Project - Console Based, Menu Driven Application

App Type	Console application - Menu driven program
Programming Language	C
Data Management	Arrays, structures and File I/O
Team / Time	Team of 2 students; 6 hours

Project Idea

Develop a console-based MCQ Exam Preparation Assistant that allows a student to maintain a question bank, practice subject-wise quizzes, view results, and track basic subject-wise progress using file-based data storage.

Functional Requirements

1. The application is designed for approximately 8-10 predefined subjects. Students may choose the subjects as per their interest. Example subjects: Aptitude, C++, Embedded C, IoT, Microcontrollers, Data Structures, Operating Systems, RTOS, Device Drivers.
2. The application should support CRUD operations for MCQs - Add, View/List, Update/Edit, and Delete.
3. Each MCQ should contain the following fields: Question Text, Four Options, Correct Answer, Short Explanation, and Subject.
4. The application should allow the student to generate a random question paper for one selected subject.
5. The student should be able to specify any valid number of questions, subject to the number of questions available for the selected subject.
6. During the quiz, questions should be presented one at a time. Revisiting already attempted questions is optional.
7. After quiz submission, the application should display the score along with the correct answer and short explanation for each question.
8. Each completed quiz attempt should be stored in a score/history file. Since this is a single-user application, all attempts are stored in the same history.
9. The application should provide basic subject-wise score analysis using the stored quiz history.

MCQ Data Fields

Field	Description	Example
Question Text	The MCQ question statement	Which keyword is used to inherit a class in Java?
Four Options	Option A, B, C and D	extends / implements / inherits / super
Correct Answer	One correct option	Extends
Short Explanation	Brief reason or concept clarification	The extends keyword is used for class inheritance.
Subject	Subject/category of the MCQ	Java

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Important Points to Note

- This is a single-user application; registration, login, logout, and user management are not required.
- This is a console-based menu-driven application.
- All application data must be stored and retrieved using files. No database is required.
- Students can choose approximately 8-10 subjects according to their interest. The exact subject list is part of the application design.
- Random paper generation must be restricted to the selected subject for that quiz attempt.
- The requested number of questions must be validated against the available question count for the selected subject.
- The application should handle invalid menu choices, invalid question counts, and invalid MCQ input gracefully.
- Quiz scores/history should persist after the application is closed and restarted.
- The program output should be well formatted and easy to understand.

Suggested File Usage

Students may decide the file structure based on their implementation. A simple design may maintain separate files for MCQs and quiz-score history. The exact file format is left to the team.

Expected Quiz Flow

1. Select a subject.
2. Enter the required number of questions.
3. Application randomly selects questions from the selected subject.
4. Display one question with four options at a time.
5. Record the selected answer and move to the next question. Revisiting is optional.
6. After the last question, submit the quiz and calculate the score.
7. Display the score and review each question with the correct answer and explanation.
8. Save the attempt and update the subject-wise analysis.

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Score Analysis - Basic Requirement

The application should use the stored quiz history to provide a simple subject-wise view of performance.

Analysis Item	Purpose	Example Display
Attempts by Subject	Shows how many quizzes were attempted	Java: 5 attempts
Best / Latest Score	Shows recent or best performance	Java: Best 9/10
Average Score	Shows basic overall performance for the subject	Java: Average 7.8/10

Minimum Expected Features

- Menu-driven navigation
- File read/write operations
- MCQ CRUD operations
- Subject-wise question management
- Random quiz generation for the selected subject
- Question-by-question attempt flow
- Score calculation and result display
- Correct answer and explanation review
- Persistent quiz history
- Basic subject-wise score analysis

Evaluation Focus

Evaluation will focus on correct functionality, menu flow, file handling, validation, code organization, and the completeness of the required features. Students are encouraged to keep the implementation simple, reliable, and well formatted within the given time.

Happy Programming!!!

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```
Windows PowerShell
=====
MCQ EXAM PREPARATION ASSISTANT
=====
MCQs: 10 | Quiz Attempts: 0

1. Manage MCQs
2. Start Random Quiz
3. View Quiz Score History
4. Subject-wise Score Analysis
5. Exit

Enter choice:
```

```
Windows PowerShell
=====
MCQ MANAGEMENT
=====
1. Add MCQ
2. View All MCQs
3. Edit MCQ
4. Delete MCQ
5. View MCQs of Given Subject
6. Back

Enter choice: |
```

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```
Windows PowerShell
=====
MCQ LIST
=====

ID: 1 | Subject: Python
Q. Which keyword is used to define a function in Python?
1. func
2. define
3. def
4. function
Correct Answer: 3
Manage MCQs -> View all MCQs

ID: 2 | Subject: Python
Q. Which data type is immutable?
1. List
2. Set
3. Dictionary
4. Tuple
Correct Answer: 4

ID: 3 | Subject: Java
Q. Which keyword is used to inherit a class in Java?
1. implements
2. extends
3. inherits
4. super
Correct Answer: 2

ID: 4 | Subject: Java
Q. Which method is the entry point of a Java application?
1. start()
```

```
Windows PowerShell
=====
ADD MCQ
=====

Subjects
1. Aptitude
2. C Programming
3. C++
4. Java
5. Python
6. Database
7. Data Structures
8. Operating Systems
9. Computer Networks
10. Web Technologies
Manage MCQ -> Add MCQ

Select subject: 1

Question: If a train covers 300 km in 5 hours, what is its speed?
Option 1: 50 km/hr
Option 2: 60 km/hr
Option 3: 70 km/hr
Option 4: 75 km/hr
Correct option (1-4): 2
Short explanation: Speed = Distance/Time = 300/5 = 60 km/hr

MCQ added successfully.

Press Enter to continue...
```

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```
Windows PowerShell
1. 50 km/hr
2. 60 km/hr
3. 70 km/hr
4. 75 km/hr
Correct Answer: 2

Manage MCQ -> Edit Question

Enter MCQ ID to edit: 11

Current question: If a train covers 300 km in 5 hours, what is its speed?
New question (press Enter to keep): If a train covers 300 km in 300 mins, what is its speed?
Option 1 (press Enter to keep): 50000 m/hr
Option 2 (press Enter to keep):
Option 3 (press Enter to keep):
Option 4 (press Enter to keep): 75000 m/hr
Correct option [2] (1-4, Enter to keep): 2
New explanation (press Enter to keep):

Current subject: Aptitude
0. Keep current subject
1. Aptitude
2. C Programming
3. C++
4. Java
5. Python
6. Database
7. Data Structures
8. Operating Systems
9. Computer Networks
10. Web Technologies
Select new subject: 0

MCQ updated successfully.
```

```
Windows PowerShell
=====
START RANDOM QUIZ
=====

Subjects
1. Aptitude
2. C Programming
3. C++
4. Java
5. Python
6. Database
7. Data Structures
8. Operating Systems
9. Computer Networks
10. Web Technologies

Main menu ->
Start the Quiz

Select subject: 1

Available questions: 3
Number of questions: 3
```

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```
Windows PowerShell
=====
APTITUDE QUIZ      Question 3 of 3
=====

What is 20% of 150?

1. 20
2. 25
3. 30
4. 35

Your answer (1-4): 3|
```

Attempting Quiz: Question by Question

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Windows PowerShell
=====
QUIZ RESULT
=====
Subject   : Aptitude
Score     : 2 / 3
Percentage: 66.67%
Quiz result (Displayed upon quiz submission)

ANSWER REVIEW
=====

1. What is 20% of 150?
   Your answer   : 1. 20
   Correct answer: 3. 30
   Explanation   : 20% of 150 is  $0.20 \times 150 = 30$ .
   Result        : Wrong

2. If 5 pens cost 100, what is the cost of 1 pen?
   Your answer   : 2. 20
   Correct answer: 2. 20
   Explanation   :  $100 \div 5 = 20$ , so one pen costs 20.
   Result        : Correct

3. If a train covers 300 km in 300 mins, what is its speed?
   Your answer   : 2. 60 km/hr
   Correct answer: 2. 60 km/hr
   Explanation   : Speed = Distance/Time =  $300/5 = 60$  km/hr
   Result        : Correct

Press Enter to continue...
```

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```
Windows PowerShell
=====
QUIZ SCORE HISTORY
=====
1. 2026-09-04 17:46:45 | Aptitude          3/3 (100.0%)
2. 2026-09-04 17:47:26 | Aptitude          2/3 (66.67%)
Press Enter to continue...|  Main menu -> View Quiz Score History
```

```
Windows PowerShell
=====
SUBJECT-WISE SCORE ANALYSIS
=====
Subject      Attempts  Average %  Best %
-----
Aptitude      2         83.34     100.0
Press Enter to continue...|  Main menu -> Subjectwise Score Analysis
```